



**BACHELOR OF INFORMATION TECHNOLOGY (HONS)**

**FINAL EXAMINATION**

**JUNE 2024**

Course : EC3247 (Cloud Architecture)  
Lecturer : Mohd Shahdi Bin Ahmad

Time: 8.00 am – 11.00 am (3 hours)  
Date: 27 September 2024

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**Instructions:**

Answer **ALL** questions in the Answer Booklet provided.

The maximum number of marks is 100.

This question paper consists of 2 printed pages.  
(excluding front cover)

**DO NOT OPEN THIS BOOKLET UNTIL YOU ARE TOLD TO DO SO**

- 1 a. Discuss the key benefits of cloud computing. (10 MARKS)  
  
b. What role does open-source software play in cloud computing? (4 MARKS)  
  
c. Describe the differences between public, private, and hybrid clouds. (6 MARKS)
- 2 a. Explain the primary layers of cloud computing architecture. (6 MARKS)  
  
b. Compare the role of middleware in PaaS with the software stack provided in IaaS. How do these layers interact to deliver cloud services? (11 MARKS)  
  
c. Define the system scalability in the context of high-performance computing (HPC). (2 MARKS)  
  
d. What law is used as a formula that predicts the theoretical speedup of a system when using multiple processors? (1 MARK)
- 3 a. Analyze how the geographic distribution of data centers impacts the performance and availability of cloud services provided by IaaS providers like Amazon Web Services. (8 MARKS)  
  
b. How does the feature of automatic scaling and load balancing in IaaS contribute to the elasticity of cloud services? (8 MARKS)  
  
c. Explain the concept of geographic distribution of data centers in IaaS providers. (2 MARKS)  
  
d. How do Service-Level Agreements (SLA) benefit users of IaaS services? (2 MARKS)
- 4 a. Analyze the impact of "data lock-in" on businesses using PaaS. What strategies can businesses adopt to mitigate this risk? (8 MARKS)  
  
b. Compare the disaster recovery capabilities of PaaS with traditional IT infrastructure. What advantages does PaaS offer in terms of fault

tolerance and disaster recovery?

- c. Analyze the security challenges in PaaS environments compared to traditional on-premises IT systems. What measures can be taken to enhance security in PaaS? **(6 MARKS)**
  
- 5 a. Analyze the impact of security challenges on the decision to migrate IT applications to the cloud. How should organizations approach these challenges to ensure a successful migration? **(8 MARKS)**
  
- b. Discuss the risks associated with cloud migration and the strategies for mitigating these risks. How does an organization balance the benefits of cloud adoption with potential risks? **(8 MARKS)**
  
- c. Explain the Public Cloud and Private Cloud. **(4 MARKS)**

**-END OF QUESTION PAPER-**